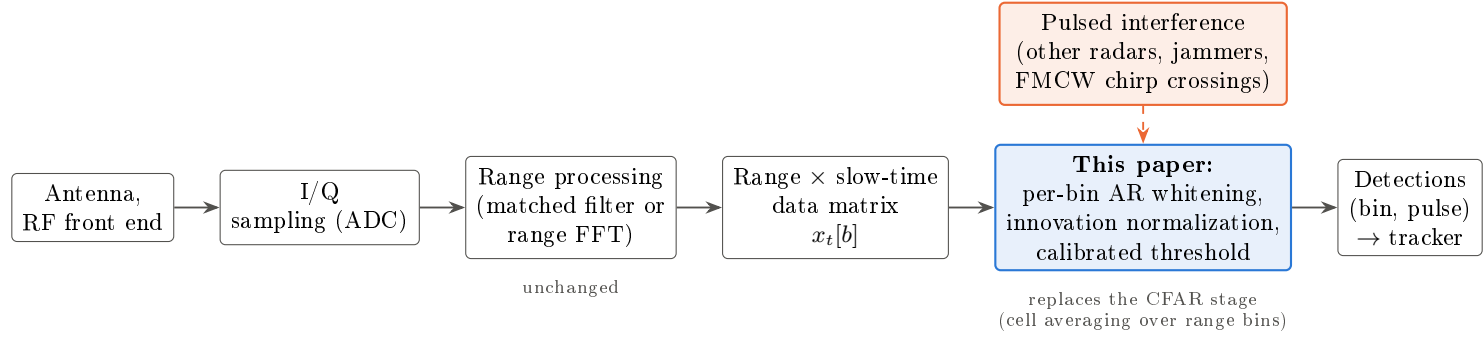


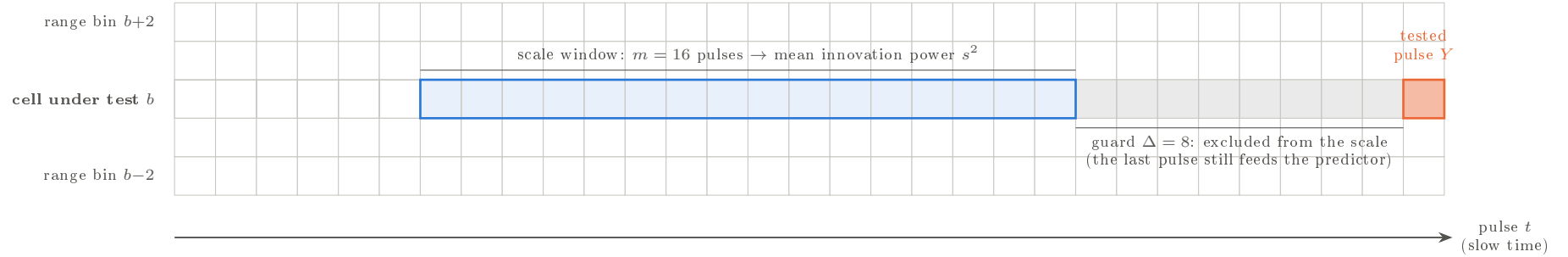
Innovation-normalized detection: what it does in the radar signal chain

Each range bin is processed on its own, along slow time (pulse to pulse). The threshold does not use neighbouring range bins.

(a) Where it fits in the receive chain



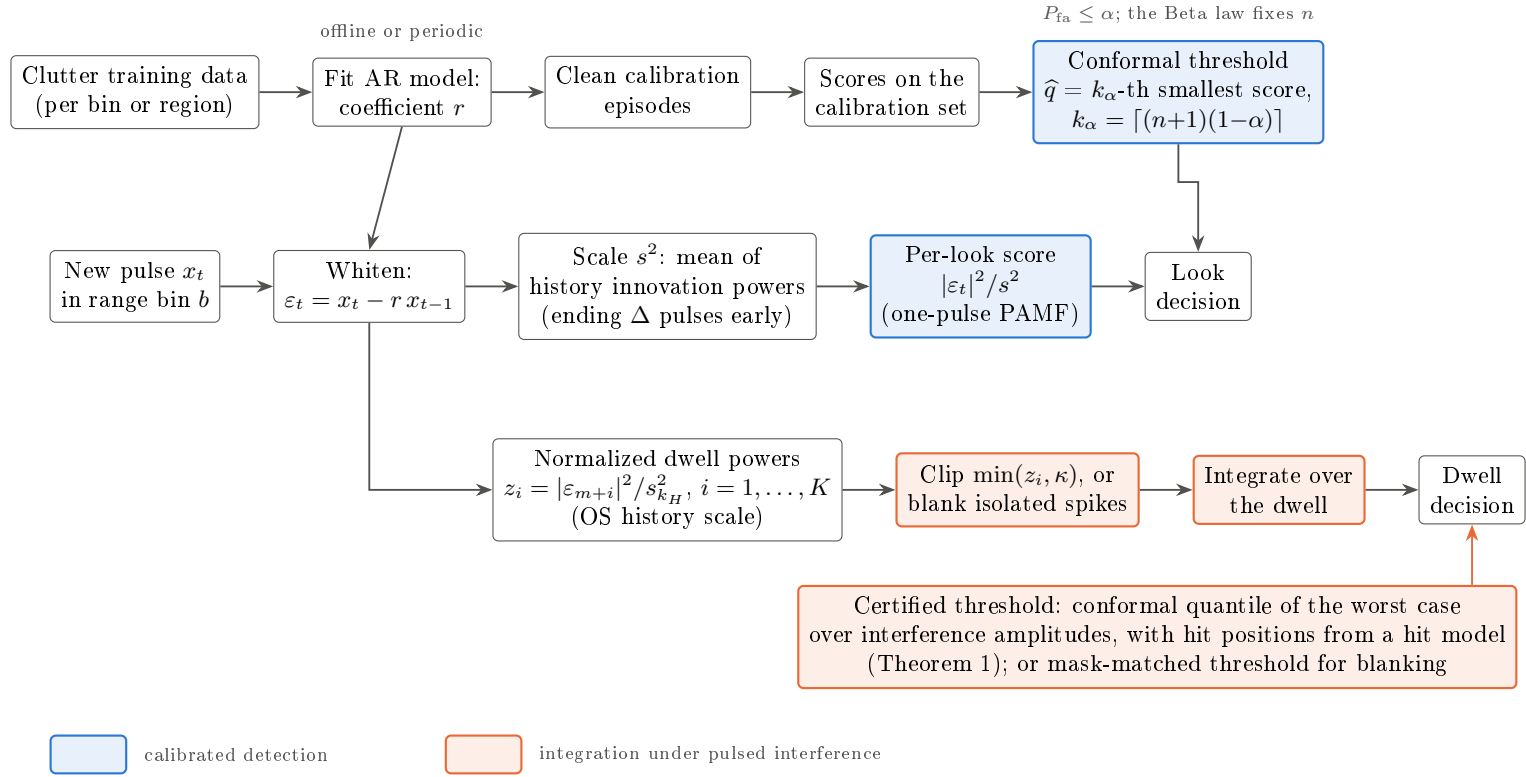
(b) What happens to one range bin: the slow-time window of one test



$$\varepsilon_t = x_t - r x_{t-1} \text{ (AR(1) innovation; } r \text{ fitted on clutter training data), } \text{ score} = |\varepsilon_Y|^2/s^2, \text{ detection if score} > \hat{q}^2 \text{ (conformal threshold).}$$

Without a guard, a target present for more than about four pulses enters the scale window and masks itself. With a guard of Δ pulses it stays visible for Δ more looks.

(c) Inside the detector



Reading the diagram. Classical CA-CFAR estimates the noise level of the cell under test from neighbouring *range* bins. That fails in sea clutter, whose strength (texture) changes from bin to bin. Here the reference is the cell's own *past*. An AR filter first removes the pulse-to-pulse correlation of the clutter (whitening), and the tested innovation is then compared with the power of the past innovations, so the clutter's local strength cancels. The threshold is the k_α -th smallest score on clean clutter (conformal), so it does not depend on a Gaussian model of the tail. A guard gap keeps a target that has just appeared out of the history. For dwell integration under pulsed interference, each normalized pulse power is clipped, or isolated spikes are blanked, before summation. The certified threshold bounds the false-alarm probability for interference of any power, provided the hit pattern is dominated by the hit model.